

FINAL EXPLANATION: CHEMISTRY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CHEMISTRY | GRADE 10 | SUB-STRAND 1.5: PERIODICITY | FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Remember the driving question: Why do elements that sit next to each other in the periodic table, like sodium and magnesium, or chlorine and argon, behave so differently, and can we use an element's position in the table to predict exactly how it will behave before we ever put it in a test tube? Use what you observed in the Lake Victoria demonstration at the Kisumu Water and Sanitation Company and in Lessons 1 to 7. Write your answers in the space under each prompt. Use electron configurations in the form 2,8,1, and use the key terms nuclear charge, shielding, ionisation energy and full outer shell. Support each claim with evidence from the lessons. Write in full sentences, and include balanced equations where asked.

Part 1: The Kisumu Puzzle - Across Period 3

At the Kisumu Water and Sanitation Company, the water purification officer dropped sodium, magnesium and aluminium into Lake Victoria water. Sodium exploded. Magnesium barely fizzed. Aluminium showed no reaction with cold water.

Use the data below (from Lessons 5 and 6) and your own knowledge of electron configurations.

Element	Atomic number	Electron configuration	First ionisation energy (kJ/mol)

(a) Sodium (2,8,1) has 11 protons and magnesium (2,8,2) has 12, but both have two full inner shells (10 inner electrons) that shield the outer electrons, so the shielding stays almost the same. The extra proton in magnesium raises the effective nuclear charge felt by the outer electrons, so they are held more tightly. The table shows this: sodium's first ionisation energy is 496 kJ/mol, but magnesium's is higher at 738 kJ/mol. Sodium's single outer electron is given up almost instantly to water, so it reacts explosively. Magnesium's outer electrons are harder to remove, so it barely fizzes in cold water.

(b) Reactivity with cold water decreases sharply across Period 3: Na (violent) > Mg (very slow) > Al (no reaction). Moving left to right, the nuclear charge increases while the shielding stays almost constant, so the outer electrons are held more strongly and it becomes harder for the element to lose electrons. The first ionisation energy rises from 496 (Na) to 738 (Mg). Aluminium's value (578) dips because its third outer electron is in the 3p sub-shell, which is slightly further out and easier to remove than magnesium's 3s electron. Even so, aluminium has a higher nuclear charge (13 protons) than sodium and must lose three electrons, so its overall tendency to lose electrons to cold water is much lower than sodium's and it shows no reaction.

(c) $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$. The solution is strongly alkaline: phenolphthalein turned it pink. The hydrogen gas ignites, which gives the orange flame.

Sodium (Na)	11	2,8,1	496
Magnesium (Mg)	12	2,8,2	738
Aluminium (Al)	13	2,8,3	578
Argon (Ar)	18	2,8,8	1521

(a) Explain why sodium and magnesium, which are neighbours separated by one proton, react so differently with cold water. Use the ideas of nuclear charge, shielding and the first ionisation energy values in the table. (b) Describe the trend in reactivity with cold water across Period 3 from sodium to aluminium, and explain it. Aluminium's first ionisation energy is lower than magnesium's; use your knowledge of electron configurations to suggest why this dip happens, and say what other factor you think explains why aluminium still shows no reaction with cold water. (c) Write a balanced equation with state symbols for the reaction of sodium with water, and state the observation from the Lake Victoria demonstration that shows the solution is alkaline.

Part 2: Going Down Group I

In Lessons 1 and 2 you saw lithium, sodium and potassium react with cold water. Lithium reacted slowly and steadily, sodium reacted faster, and potassium reacted most violently with a lilac flame.

(a) Write the electron configurations of Li, Na and K. (b) Explain why potassium is more reactive than sodium even though potassium has many more protons. Your explanation must refer to the number of electron shells, atomic radius, shielding and ionisation energy. (c) Name the two products of the reaction of potassium with water and describe how you would prove that one of them is a gas.

(a) Li is 2,1; Na is 2,8,1; K is 2,8,8,1. Each has one outer electron.
 (b) Going down Group I, each element has one more electron shell, so the atomic radius increases. The outer electron is further from the nucleus and is shielded by more inner electrons (10 in sodium, 18 in potassium). The extra protons in potassium are outweighed by this extra shielding and distance, so the outer electron feels a weaker pull from the nucleus. The first ionisation energy is therefore lower for potassium than for sodium. Its outer electron is lost more easily, so potassium is more reactive and reacts more violently ($\text{Li} < \text{Na} < \text{K}$).
 (c) The products are potassium hydroxide (KOH) and hydrogen gas (H_2): $2\text{K(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{KOH(aq)} + \text{H}_2\text{(g)}$. To prove the gas is hydrogen, collect it over water and hold a lighted splint to it; hydrogen gives a squeaky pop. The alkaline solution is also shown by a high pH on universal indicator.

Part 3: Two Families, Opposite Trends

In Lesson 4 you studied Group VII, the halogens: chlorine (Cl_2 , yellow-green gas), bromine (Br_2 , orange-brown liquid) and iodine (I_2 , grey-black solid). Chlorine water displaced bromide ions from potassium bromide solution, and bromine water displaced iodide ions from potassium iodide solution.

(a) Write a word or balanced equation for the displacement reaction of chlorine water with potassium bromide solution. (b) Compare how reactivity

(a) $\text{Cl}_2\text{(aq)} + 2\text{KBr(aq)} \rightarrow 2\text{KCl(aq)} + \text{Br}_2\text{(aq)}$. Chlorine is more reactive than bromine, so it displaces bromine. The cyclohexane layer turns orange-brown.
 (b) Reactivity increases down Group I ($\text{Li} < \text{Na} < \text{K}$) but decreases down Group VII ($\text{Cl} > \text{Br} > \text{I}$).
 (c) In Group I, atoms react by losing an outer electron. More shells mean more shielding and a larger atom, so the electron is lost more easily (lower ionisation energy) and reactivity increases. In Group VII, atoms react by gaining an electron. More shells mean a larger atom with more shielding, so the nucleus attracts an incoming electron less strongly (lower electron affinity) and it is harder to gain an electron, so reactivity decreases. The two trends are mirror images: the same cause, more shells and more shielding, makes losing electrons easier but gaining electrons harder.
 (d) Chlorine has the smallest atomic radius and the highest electron affinity of the three, so it is the strongest oxidising agent. It reacts with water to form hypochlorous acid, a bleaching and disinfecting solution (it bleached the kanga fabric within 60 seconds), which makes it effective for disinfecting Lake Victoria water.

changes down Group I with how it changes down Group VII. (c) Explain why the two trends go in opposite directions, even though both are caused by more electron shells. (d) Explain why the Kisumu water officer uses chlorine rather than bromine or iodine to treat Lake Victoria water.

Part 4: The Inert Shield

The officer transported the sodium in a sealed ampoule of argon gas. Argon sits right next to the very reactive chlorine, yet it reacts with nothing. Use the table in Part 1 and your knowledge of electron configurations.

(a) Compare the first ionisation energy of argon with that of sodium. How many times larger is the argon value, approximately? (b) Explain why argon is completely inert, referring to its outer shell. (c) Explain why argon, rather than ordinary air, was needed to protect the sodium.

(a) Argon's first ionisation energy (1521 kJ/mol) is about three times larger than sodium's (496 kJ/mol), since $3 \times 496 = 1488$, and 1521 is slightly more than that.

(b) Argon is 2,8,8, so its outer shell is completely full (an octet). Its electrons are held strongly and are very difficult to remove, and a full shell has no tendency to gain or share electrons either. This is why argon does not react under ordinary conditions. Chlorine (2,8,7) is one electron short of a full shell, so it readily gains an electron and is very reactive.

(c) Sodium's single outer electron is lost very easily (low ionisation energy), so sodium reacts even with trace moisture in air. Argon is inert and will not react with sodium or with anything else, so it gives a completely safe shielding atmosphere for the sodium during transport.

Part 5: Final Explanation - Predicting from Position

Now answer the driving question in a single connected explanation: Why do neighbouring elements such as sodium and magnesium, or chlorine and argon, behave so differently, and can we use an element's position in the periodic table to predict how it

An element's position tells us its electron configuration. The period number tells us how many electron shells it has, and the group number tells us how many outer electrons it has (for example Na is Period 3, Group I: 2,8,1). The electron configuration controls how strongly the outer electrons are held, which controls how an element reacts, so neighbours can behave very differently.

Across Period 3, each step adds one proton while the shielding stays almost the same, so the effective nuclear charge rises and the outer electrons are held more tightly. Ionisation energy generally increases (Na 496, Mg 738 kJ/mol), so reactivity with cold water falls: Na (2,8,1) explodes in Lake Victoria water, Mg (2,8,2) barely fizzes and Al shows no reaction. Argon (2,8,8) has a full outer shell and is inert, which is why it can shield sodium. Down Group I, each element has an extra shell, so the outer electron is further away and more shielded; ionisation energy falls and reactivity rises ($\text{Li} < \text{Na} < \text{K}$). In Group VII the same extra shielding

will behave before we put it in a test tube? Your answer must: (1) explain what an element's group and period tell you about its electron configuration; (2) explain the Group I down-group trend and the Period 3 across-period trend using shielding, nuclear charge and ionisation energy; (3) use at least one example of a Kenyan application of an element and link it to a periodic property; (4) give a prediction for how a new element would behave based on its position, and justify it. Choose either calcium (Group II, 2,8,8,2) compared with magnesium, or rubidium (below potassium in Group I).	makes electron gain harder, so reactivity falls ($\text{Cl} > \text{Br} > \text{I}$). Applications follow from these properties. Chlorine, which needs only one electron to fill its outer shell, is a strong oxidising agent and disinfects water at the Kisumu plant. Argon's full shell makes it an inert welding shield. Prediction: calcium (2,8,8,2) has an extra shell compared with magnesium (2,8,2). The extra shielding lowers its ionisation energy, so I predict that calcium reacts faster with cold water than magnesium, producing hydrogen and an alkaline solution of calcium hydroxide, but less violently than sodium or potassium because it must lose two electrons. This agrees with what we observed in Lesson 3. So yes: position lets us predict behaviour before we ever use a test tube.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of electron configuration and position	Correctly writes electron configurations and clearly links group and period to outer electrons and shells in every part.	Writes mostly correct configurations and links position to electrons in most parts, with minor gaps.	Configurations are missing or incorrect, and position is not clearly linked to electrons.
Explanation of trends using nuclear charge, shielding and ionisation energy	Accurately explains the Group I, Period 3 and Group VII trends, using all three ideas and showing why Group I and Group VII trends are opposite.	Explains most trends with two of the key ideas, with some imprecision.	Describes trends without a causal explanation, or uses ideas incorrectly.
Use of evidence and data	Cites specific observations and values (such as 496, 738 and 1521 kJ/mol, lilac flame, kanga bleaching) accurately and consistently.	Cites some evidence accurately, though not always specifically.	Little or no evidence cited, or the evidence is inaccurate.
Equations and chemical accuracy	Balanced equations with correct state symbols, correct products and correct tests.	Equations mostly correct, with minor errors in balancing or state symbols.	Equations incorrect or missing.
Application, prediction and	Gives a justified prediction and a Kenyan	Gives a reasonable prediction and	Prediction or application is missing or

connection to the phenomenon	application, and returns to the Kisumu phenomenon to answer the driving question completely.	application with a partial link to the phenomenon.	unjustified, with no clear link back to the phenomenon.
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